Clinical Study Summary Report (CSSR) Document Status: [Draft] | **Version:** [1.0] | **Date:** [06-Apr-2026]

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1. Administrative & Study Identification

Official Title: A 2-year randomized, 3-arm, double-blind, non-inferiority study comparing the efficacy and safety of ofatumumab and siponimod versus fingolimod in pediatric patients with multiple sclerosis followed by an open-label extension. **Brief Title:** Efficacy and Safety of Ofatumumab and Siponimod Compared to Fingolimod in Pediatric Patients With Multiple Sclerosis **Short Title/Acronym:** NEOS **Protocol Number:** CBAF312D2301 **NCT Number:** NCT04926818 **Posting ID:** 850098328788839706 **Sponsor/Company Name:** Novartis **Investigational Product:** Ofatumumab, Siponimod, and Fingolimod (Active Control) **Phase of Development:** PHASE3 **Trial Status:** ACTIVE_NOT_RECRUITING **Disease Area:** Multiple Sclerosis (MS) **Medical Monitor:** Claus-Peter Danzer, Novartis Pharma AG (claus-peter.danzer@novartis.com)

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2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate the efficacy and safety of ofatumumab and siponimod compared to fingolimod in pediatric patients (aged 10-17 years) with multiple sclerosis, primarily by evaluating if the new treatments are non-inferior to fingolimod regarding the Annualized Relapse Rate (ARR) in target pediatric participants. * **Secondary Objectives:** To evaluate long-term safety, tolerability, and additional efficacy outcomes (like MRI lesion reduction and disability progression) during the 24-month Core Part and the subsequent 60-month extension.

2.2 Background & Rationale While fingolimod works well and is an approved medication for MS in children ≥ 10 years, there is a critical need for advanced therapeutic options for pediatric-onset multiple sclerosis (POMS). Ofatumumab (a fully human anti-CD20 monoclonal antibody) and siponimod (an S1P modulator) have demonstrated significant efficacy in adults. The NEOS study aims to extrapolate and confirm that these agents are safe and beneficial in reducing clinical attacks/relapses and MRI lesions in the vulnerable pediatric MS population.

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3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Active Comparator (Fingolimod) * **Blinding:** Double-blind, triple-dummy (during the 24-month Core Part) * **Randomization:** Randomized, 3-arm

3.2 Planned Enrollment & Duration * **Targeted Enrollment:** 129 pediatric patients (target includes at least 120 participants with multiple sclerosis, with subgroups of at least 5 participants with body weight ≤ 40 kg and at least 5 participants with age 10-12 years in each of the ofatumumab and siponimod arms). * **Number of Sites:** Multicenter, ~51 global locations * **Estimated Study Start:** 05-Oct-2021 * **Estimated Primary Completion:** 06-Mar-2031 (Study estimated full completion: 23-Dec-2031) * **Total Duration:** Up to 7 years.

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Between 10 to <18 years of age (i.e., have not yet had their 18th birthday) at randomization. 2. Diagnosis of multiple sclerosis. 3. Expanded Disability Status Scale (EDSS) score of 0 to 5.5, inclusive. 4. At least one MS relapse/attack during the previous year, OR two MS relapses in the previous two years prior, OR evidence of one or more new T2 lesions within 12 months.

4.2 Exclusion Criteria 1. Participants with progressive MS. 2. Participants with an active, chronic disease of the immune system other than MS. 3. Participants meeting the definition of Acute Disseminated Encephalomyelitis (ADEM). 4. Participants with severe cardiac disease or significant findings on the screening ECG. 5. Participants with severe renal insufficiency.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** * *Ofatumumab:* 20 mg loading dose at Day 1, Day 7, and Day 14, then injected every 4 or 6 weeks depending on body weight. * *Siponimod:* 0.5 mg, 1 mg, or 2 mg once daily (following a titration period from Day 1 to 6). * *Fingolimod (Control):* 0.5 mg or 0.25 mg once daily, dependent on body weight (ATC Code: L04AA27; Trade Names: Gilenya, Tascenso). * **Route of Administration:** Subcutaneous (Ofatumumab via autoinjector) and Oral (Siponimod tablets, Fingolimod capsules). * **Duration of Treatment:** 24-month Core Part, followed by a 60-month (5-year) open-label Extension Part (except for the first 12 weeks transition which will remain double-blind), for a total duration of up to 7 years. Minimum 6-month follow-up period for all participants.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Standard symptomatic therapies for MS relapses (e.g., antipyretics or NSAIDs for injection-site reactions). * **Prohibited:** Any other concurrent disease-modifying therapies (DMTs) for MS during the triple-dummy blinded phase.

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6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent/ Assent, Medical History, Diagnosis Confirmation, EDSS score, MRI | | **Baseline** | Day 1 | Randomization, First Dose administration (Triple-Dummy), Titration Initiation | | **Interim** | Core Part (24 Months) | Efficacy/Safety monitoring, Relapse Assessment, MRI tracking, EDSS evaluations | | **Extension**| Up to Month 84 | Open-label treatment (initial 12-week transition double-blind), long-term safety | | **End of Study** | Month 84 (Year 7) | Final EDSS/ MRI, End of Extension Part, minimum 6-month safety follow-up |

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Annualized Relapse Rate (ARR) in target pediatric participants. The ARR is defined as the average number of confirmed relapses per year (total number of confirmed relapses divided by the total days in the study multiplied by 365.25). * **Measurement Method:** Clinical assessment of documented MS relapses evaluated over the Core Part.

7.2 Secondary & Exploratory Endpoints * Annualized relapse rate (ARR) as compared to historical interferon β-1a data. * Annualized T2 lesion rate. * Neurofilament light chain (NfL) concentrations. * Confirmed disability progression measured by Expanded Disability Status Scale (EDSS). * Long-term safety and tolerability assessment (including AE tracking) throughout the study.

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8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Standard collection evaluating clinical and laboratory markers, with specific monitoring for infection, hepatic changes, and potential injection site reactions.
- **Serious Adverse Events (SAEs):** Specific monitoring for cardiac issues (e.g., bradycardia associated with S1P modulators) requiring 24-hour reporting.
- **Data Monitoring Committee (DMC):** External and internal pediatric safety monitoring panels per Novartis guidelines.

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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) for non-inferiority efficacy assessments, Safety Set for adverse events.
- **Sample Size Justification:** Targeted enrollment $N = 129$, providing sufficient power for a combined Bayesian non-

inferiority design leveraging historical adult data (e.g., TRANSFORMS) to establish pediatric outcomes.
- **Handling of Missing Data:** Utilizing a negative binomial model for relapse rates with age-by-treatment interactions, implementing Multiple Imputation/LOCF methodology for MRI and clinical data withdrawals.

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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
- **Monitoring Plan:** Routine onsite and remote monitoring visits for triple-dummy blinding assurance.
- **Audit Trail:** Strict documentation of the transition from the double-blind 24-month Core Part to the 60-month open-label Extension Part via validated eCRFs.

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11. Study Identifiers & Governance

- **Primary ID:** CBAF312D2301
- **Registry Number (NCT):** NCT04926818
- **Posting ID:** 850098328788839706
- **Sponsor/Lead Organization:** Novartis
- **Study Title:** Efficacy and Safety of Ofatumumab and Siponimod Compared to Fingolimod in Pediatric Patients With Multiple Sclerosis
- **Official Regulatory Title:** A 2-year randomized, 3-arm, double-blind, non-inferiority study comparing the efficacy and safety of ofatumumab and siponimod versus fingolimod in pediatric patients with multiple sclerosis followed by an open-label extension.
- **Trial Status:** ACTIVE_NOT_RECRUITING

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12. Clinical Framework (Multiple Sclerosis Specific)

- **Primary Condition:** Pediatric-Onset Multiple Sclerosis (POMS) / Preferred UMLS Name: Multiple Sclerosis
- **Diagnosis Threshold:** Diagnosis of MS with EDSS score of 0 to 5.5 and ≥ 1 relapse in the past year or 2 in previous two years, or new T2 lesions.
- **Medical Methodology:** Disease-Modifying Therapy (DMT) implementation
- **Optimization Target:** Reduction in Annualized Relapse Rate (ARR) and suppression of active MRI lesions

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13. Study Procedures & Methodology

- **Management Pathway:** Standard clinical pathway for POMS utilizing step-therapy and high-efficacy DMTs.
- **Education Protocol (HCP):** Specialized training on maintaining the triple-dummy blinding and pediatric MS assessment scales.
- **Education Protocol (Patient):** Subcutaneous autoinjector training for ofatumumab and adherence counseling for daily oral agents.
- **Quality Audit Mechanism:** Medication possession ratio monitoring and EDSS/MRI standardized central reading.

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14. Observation & Follow-Up Schedule

- **Total Duration:** Up to 84 months (24 months Core + 60 months Extension)
- **Onsite Visit Cadence:** Baseline, Day 7, Day 14, followed by routine monthly/quarterly clinic visits as defined by treatment schedules.
- **Remote Monitoring Frequency:** Intermittent remote assessments for ongoing symptom management.
- **Checklist Review Interval:** Regular interval tracking (e.g., 6-12 months) of MRI milestones and clinical relapses.

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]